DATABASE RELATIONSHIP ANALYSIS USING JOINS

1.OBJECTIVES :

- To combine Customer, Product, Order, and Payment tables.
- To implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
- To retrieve complete order details.
- To display customer purchase history.
- To generate multi-table reports.

2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.

2.1 Customer and Orders

```
SQL> SELECT C.ID,
  2         C.NAME,
  3         O.Order_ID,
  4         O.Order_Date
  5  FROM Customer C
  6  INNER JOIN Orders O
  7  ON C.ID = O.Customer_ID;

        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
         1 Ramesh                           1001 01-SEP-26
         2 Khilan                           1002 03-SEP-26
         3 Koushik                          1003 05-SEP-26
         4 Chaitali                         1004 07-SEP-26
         5 Hardik                           1005 10-SEP-26
         1 Ramesh                           1006 12-SEP-26
         3 Koushik                          1007 15-SEP-26
         5 Hardik                           1008 18-SEP-26

8 rows selected.
```

This displays customers who have placed orders.

2.2 Orders and Payment

```
SQL> SELECT O.Order_ID,
  2         O.Order_Date,
  3         P.Payment_ID,
  4         P.Payment_Method,
  5         P.Payment_Status,
  6         P.Amount
  7  FROM Orders O
  8  INNER JOIN Payment P
  9  ON O.Order_ID = P.Order_ID;

  ORDER_ID ORDER_DAT PAYMENT_ID PAYMENT_METHOD
---------- --------- ---------- ------------------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
      1001 01-SEP-26       1001 UPI
Successful                 2500

      1002 03-SEP-26       1002 Cash
Successful                 1499

      1003 05-SEP-26       1003 UPI
Successful                 2598


  ORDER_ID ORDER_DAT PAYMENT_ID PAYMENT_METHOD
---------- --------- ---------- ------------------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
      1004 07-SEP-26       1004 Card
Successful                 1999

      1005 10-SEP-26       1005 UPI
Pending                     450

      1006 12-SEP-26       1006 Cash
Successful                 1299


6 rows selected.
```

This displays order and corresponding payment details

2.3 Product and Order_Details

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         OD.Order_ID,
  4         OD.Quantity
  5  FROM Product P
  6  INNER JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID;

PRODUCT_ID PRODUCT_NAME                                       ORDER_ID
---------- -------------------------------------------------- ----------
  QUANTITY
----------
       101 Laptop                                                    201
         3

       102 Smartphone                                                201
         1

       103 Headphones                                                202
         2


PRODUCT_ID PRODUCT_NAME                                       ORDER_ID
---------- -------------------------------------------------- ----------
  QUANTITY
----------
       104 Keyboard                                                  202
         1

       105 Mouse                                                     203
         2

       106 Monitor                                                   203
         1


6 rows selected.
```

This displays products that have been included in orders.

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table. If there is no match, NULL values are returned.

3.1 Display All Customers and Their Orders

```
SQL> SELECT C.ID, C.NAME,
  2         O.ORDER_ID, O.ORDER_DATE
  3  FROM Customer C
  4  LEFT JOIN Orders O
  5  ON C.ID = O.CUSTOMER_ID;

        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
         1 Ramesh                               1001 01-SEP-26
         2 Khilan                               1002 03-SEP-26
         3 Koushik                              1003 05-SEP-26
         4 Chaitali                             1004 07-SEP-26
         5 Hardik                               1005 10-SEP-26
         1 Ramesh                               1006 12-SEP-26
         3 Koushik                              1007 15-SEP-26
         5 Hardik                               1008 18-SEP-26

8 rows selected.
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table. If there is no match, NULL values are returned.

4.1 Display All Orders and Customer Details

```
SQL> SELECT O.ORDER_ID, O.CUSTOMER_ID, O.ORDER_DATE,
  2         C.ID, C.NAME
  3  FROM Customer C
  4  RIGHT JOIN Orders O
  5  ON C.ID = O.CUSTOMER_ID;

  ORDER_ID CUSTOMER_ID ORDER_DAT         ID NAME
---------- ----------- --------- ---------- ------------------------------
      1001           1 01-SEP-26          1 Ramesh
      1006           1 12-SEP-26          1 Ramesh
      1002           2 03-SEP-26          2 Khilan
      1003           3 05-SEP-26          3 Koushik
      1007           3 15-SEP-26          3 Koushik
      1004           4 07-SEP-26          4 Chaitali
      1005           5 10-SEP-26          5 Hardik
      1008           5 18-SEP-26          5 Hardik

8 rows selected.
```

5. Complete Order Details

```
SQL> SELECT C.ID, C.NAME,
  2         O.ORDER_ID, O.ORDER_DATE, O.ORDER_STATUS, O.TOTAL_AMOUNT,
  3         P.PAYMENT_ID, P.PAYMENT_METHOD, P.PAYMENT_STATUS, P.AMOUNT
  4  FROM Customer C
  5  INNER JOIN Orders O
  6  ON C.ID = O.CUSTOMER_ID
  7  INNER JOIN Payment P
  8  ON O.ORDER_ID = P.ORDER_ID
  9  ORDER BY O.ORDER_ID;

        ID NAME                             ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         TOTAL_AMOUNT PAYMENT_ID PAYMENT_METHOD
-------------------- ------------ ---------- ------------------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
         1 Ramesh                               1001 01-SEP-26
Placed                     2098       1001 UPI
Successful                 2500

         2 Khilan                               1002 03-SEP-26
Confirmed                  1499       1002 Cash
Successful                 1499
```

```
        ID NAME                             ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         TOTAL_AMOUNT PAYMENT_ID PAYMENT_METHOD
-------------------- ------------ ---------- ------------------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------

         3 Koushik                              1003 05-SEP-26
Shipped                    2598       1003 UPI
Successful                 2598

         4 Chaitali                             1004 07-SEP-26
Delivered                  1999       1004 Card

        ID NAME                             ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         TOTAL_AMOUNT PAYMENT_ID PAYMENT_METHOD
-------------------- ------------ ---------- ------------------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
Successful                 1999

         5 Hardik                               1005 10-SEP-26
Placed                      450       1005 UPI
Pending                     450

         1 Ramesh                               1006 12-SEP-26

        ID NAME                             ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         TOTAL_AMOUNT PAYMENT_ID PAYMENT_METHOD
-------------------- ------------ ---------- ------------------------------
PAYMENT_STATUS           AMOUNT
-------------------- ----------
Confirmed                  1299       1006 Cash
Successful                 1299


6 rows selected.
```

6. Customer Purchase History

```
SQL> SELECT C.ID, C.NAME,
  2         O.ORDER_ID, O.ORDER_DATE,
  3         O.ORDER_STATUS, O.TOTAL_AMOUNT
  4  FROM Customer C
  5  INNER JOIN Orders O
  6  ON C.ID = O.CUSTOMER_ID
  7  ORDER BY C.ID, O.ORDER_DATE;

        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         TOTAL_AMOUNT
-------------------- ------------
         1 Ramesh                             1001 01-SEP-26
Placed                       2098

         1 Ramesh                             1006 12-SEP-26
Confirmed                    1299

         2 Khilan                             1002 03-SEP-26
Confirmed                    1499


        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         TOTAL_AMOUNT
-------------------- ------------
         3 Koushik                            1003 05-SEP-26
Shipped                      2598

         3 Koushik                            1007 15-SEP-26
Shipped                      1999

         4 Chaitali                           1004 07-SEP-26
Delivered                    1999
```

```
        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         TOTAL_AMOUNT
-------------------- ------------
         5 Hardik                             1005 10-SEP-26
Placed                        450

         5 Hardik                             1008 18-SEP-26
Delivered                    1049


8 rows selected.
```

8. Multi-Table Business Report

```
SQL> SELECT C.ID, C.NAME,
  2         O.ORDER_ID, O.ORDER_DATE, O.ORDER_STATUS,
  3         P.PAYMENT_METHOD, P.PAYMENT_STATUS, P.AMOUNT
  4  FROM Customer C
  5  INNER JOIN Orders O
  6  ON C.ID = O.CUSTOMER_ID
  7  INNER JOIN Payment P
  8  ON O.ORDER_ID = P.ORDER_ID
  9  ORDER BY O.ORDER_ID;

        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         PAYMENT_METHOD                 PAYMENT_STATUS
-------------------- ------------------------------ --------------------
    AMOUNT
----------
         1 Ramesh                                1001 01-SEP-26
Placed               UPI                            Successful
      2500

         2 Khilan                                1002 03-SEP-26
Confirmed            Cash                           Successful
      1499

        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         PAYMENT_METHOD                 PAYMENT_STATUS
-------------------- ------------------------------ --------------------
    AMOUNT
----------

         3 Koushik                               1003 05-SEP-26
Shipped              UPI                            Successful
      2598

         4 Chaitali                              1004 07-SEP-26
Delivered            Card                           Successful

        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         PAYMENT_METHOD                 PAYMENT_STATUS
-------------------- ------------------------------ --------------------
    AMOUNT
----------
      1999

         5 Hardik                                1005 10-SEP-26
Placed               UPI                            Pending
       450

         1 Ramesh                                1006 12-SEP-26

        ID NAME                           ORDER_ID ORDER_DAT
---------- ------------------------------ ---------- ---------
ORDER_STATUS         PAYMENT_METHOD                 PAYMENT_STATUS
-------------------- ------------------------------ --------------------
    AMOUNT
----------
Confirmed            Cash                           Successful
      1299


6 rows selected.
```

9. Payment Transaction Report

```
SQL> SELECT P.PAYMENT_ID, P.ORDER_ID,
  2         P.PAYMENT_METHOD, P.PAYMENT_STATUS,
  3         P.PAYMENT_DATE, P.AMOUNT,
  4         O.CUSTOMER_ID, O.ORDER_STATUS
  5  FROM Payment P
  6  INNER JOIN Orders O
  7  ON P.ORDER_ID = O.ORDER_ID
  8  ORDER BY P.PAYMENT_ID;

PAYMENT_ID   ORDER_ID PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- ------------------------------ --------------------
PAYMENT_D     AMOUNT CUSTOMER_ID ORDER_STATUS
--------- ---------- ----------- --------------------
      1001       1001 UPI                            Successful
27-OCT-26       2500           1 Placed

      1002       1002 Cash                           Successful
27-OCT-26       1499           2 Confirmed

      1003       1003 UPI                            Successful
27-OCT-26       2598           3 Shipped


PAYMENT_ID   ORDER_ID PAYMENT_METHOD                 PAYMENT_STATUS
---------- ---------- ------------------------------ --------------------
PAYMENT_D     AMOUNT CUSTOMER_ID ORDER_STATUS
--------- ---------- ----------- --------------------
      1004       1004 Card                           Successful
27-OCT-26       1999           4 Delivered

      1005       1005 UPI                            Pending
27-OCT-26        450           5 Placed

      1006       1006 Cash                           Successful
27-OCT-26       1299           1 Confirmed


6 rows selected.
```